

Collections Hackathon

*Unify the data. Ask it anything. **Power the next best action.***

Challenge brief for participating teams

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COLLECTIONS

Helping customers who fall behind get back on track



What collections does

Contacts customers who have missed card or loan payments, agrees a plan and recovers what is owed.



Why it's hard

Each customer's story is scattered across many systems, from account data to call recordings.



Where AI helps

Giving agents the right context and choosing the right next action, fast and fairly.

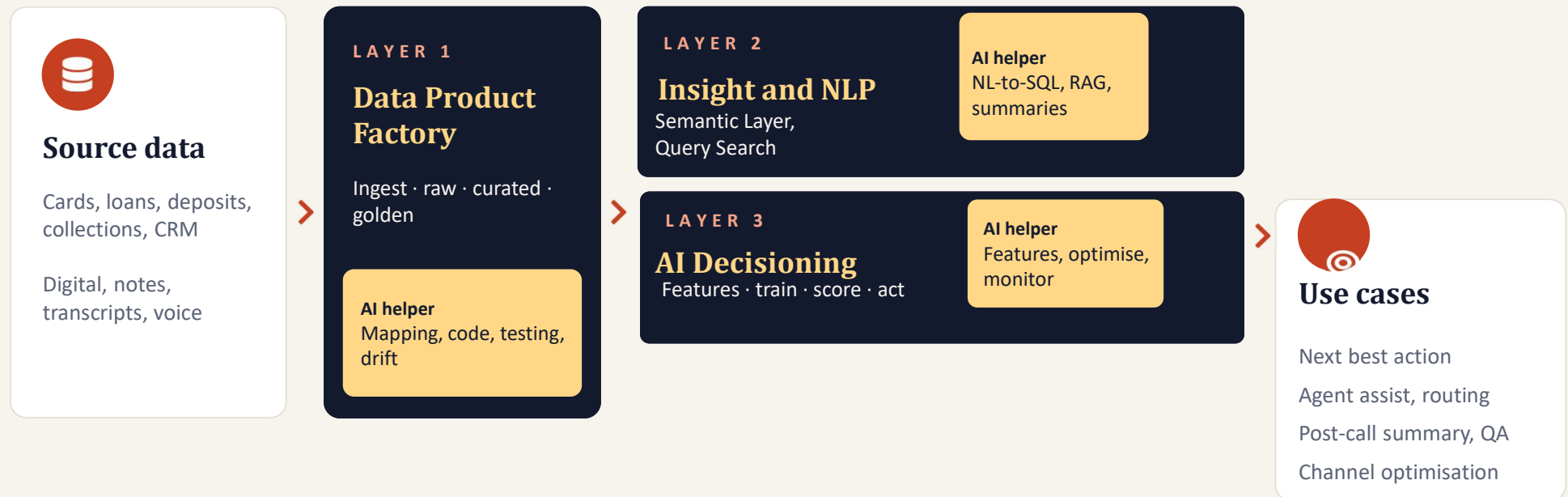
THE CHALLENGE

Collections data and insights are fragmented across cards, loans, deposits, collections systems, CRM, digital interactions, agent notes, transcripts, voice recordings and external data.

Design a student-ready solution for the Collections function that unifies enterprise and interaction data, enables natural-language access, and powers operational AI use cases.

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Three layers, from raw data to decisions



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Governance runs through every layer: security, access control, monitoring and a human in the loop.

KEY CONCEPT

What we mean by a data product

A **reusable, trusted, governed and business-owned** data asset that serves many use cases for one or more consumers.

It is

- ✓ Built for a clear business purpose
- ✓ Owned and managed like a product
- ✓ Documented with metadata and definitions
- ✓ Held to agreed quality rules
- ✓ Used by people, apps, analytics and AI

It isn't

- ✗ A temporary working file
- ✗ A one-off extract
- ✗ A personal spreadsheet
- ✗ An intermediate engineering table
- ✗ Built for a single use case

Turn fragmented sources into a Collections 360

01 Raw datasets



02 Curated (silver)



03 Golden C360



04 Feature-ready products



Where AI should help

- Source-to-target mapping suggestions
- SQL transformation code generation
- Schema drift detection
- Anomaly detection in pipelines



You must deliver

- C360 schema and pipeline code
- Customer ID matching across sources
- One data contract with quality rules
- A data quality report

EXAMPLE [Just for reference] ·

What a Collections 360 record looks like

c360.collections.bank / customer / G-004817

Collections 360 Golden record

RM **Ryan Mitchell** · G-004817 · Hourly wage · Windsor, ON · customer since 2016

Risk: High 0.71 · Bucket 31-60 · Consent: Call 8am-8pm ET · SMS · Email · EN · Handship signal

PRODUCT HOLDINGS

Product	Account	Balance	Overdue	DPD	Status
Credit card (Visa)	**4417	\$8,430	\$412	38	31-60
Personal loan	**0921	\$18,650	\$1,160	34	2 payments missed
Chequing	**7730	\$214	—	—	No payroll deposit since 14 Aug
Credit bureau (external)	score	648	↓ 42 in 90 days	—	2 new inquiries

CONTACT TIMELINE

25 Sep	Call	New PTP \$800 by 2 Oct · agent A. Roy · transcript #8840
18 Sep	SMS	Payment reminder · delivered, no response
15 Sep	System	PTP broken · \$1,000 not received
12 Sep	Call	"Shifts cut at the plant since August" · PTP \$1,000 by 15 Sep · #8812
05 Sep	App	Push notification opened · no payment

IDENTITY RESOLUTION & LINEAGE
4 source IDs → 1 golden ID · match confidence 0.97
cards: C-88213 · loans: L-5530921
crm: RYM-1142 · deposits: CHQ-7730
Lineage: 9 sources · cards, loans, deposits, collections, CRM, contacts, notes, transcripts, bureau

1 One customer, one ID

Records from 9 source systems resolve to one golden ID, with the match confidence shown.

2 Trusted, not just joined

Every field carries lineage, a refresh time and the quality checks behind it.

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Collections Hackathon · IIIT [Campus]

Illustrative mock-up · synthetic data

08

Let collections teams ask questions in plain English



Which customers are most likely to break a promise to pay?



Which strategy worked best for unsecured loans in the last 90 days?



What were the top drivers of roll-rate deterioration this week?



Where AI should help

- NL-to-SQL over curated and golden data
- RAG over agent notes and transcripts
- Summaries and insight generation



You must deliver

- A working demo on [] benchmark questions
- The SQL or source shown for every answer
- Refusal of questions outside allowed data

EXAMPLE · [Just for reference]

Ask a question, get an answer with its working

Collections Ask Search your collections data in plain English

Which customers are most likely to break a promise to pay this week

Understood as: open promises to pay due 28 Sep - 4 Oct sort by break probability ↓

8 open promises most likely to break - \$12,990 at risk

Customer	Product	PTP amount	Due	Break prob.
G-004817 R. Mitchell	PL + Card	\$800	2 Oct	0.81
G-011203 S. Tremblay	Card	\$2,250	30 Sep	0.78
G-007741 A. Nguyen	PL	\$1,900	1 Oct	0.74
G-020915 P. Gagnon	Card	\$640	29 Sep	0.69
G-003318 M. Roy	Auto	\$3,100	3 Oct	0.66
G-016672 K. MacDonald	PL	\$1,450	4 Oct	0.63
G-009024 V. Chen	Card	\$1,600	2 Oct	0.61
G-012890 T. Campbell	PL	\$1,250	30 Sep	0.60

HOW I GOT THIS

```
SELECT c.golden_id, c.name, p.product, p.ptp_amount, p.ptp_due_date,
       s.score AS break_prob, s.top_reason_1_text
FROM gold.promises_to_pay p JOIN gold.model_scores s
ON s.ptp_id = p.ptp_id AND s.model_name = 'ptp_break' AND s.score_date = '2026-09-28'
JOIN gold.c360_customer c USING(golden_id)
WHERE p.ptp_status = 'open' AND p.ptp_due_date BETWEEN '2026-09-28' AND '2026-10-04'
ORDER BY break_prob DESC LIMIT 8;
```

Why is G-004817 high risk? Split by agent team
Compare with last week Which have hardship flags?

1 Show what it understood

The question becomes editable search tokens, so a misread filter is easy to fix.

2 Show the working

Every answer carries its SQL and the data products it used.

3 Keep the analysis going

An LLM insight plus suggested follow-up questions.

Pick use case and make it work



Next best action

Recommends the right treatment, channel, and timing for every customer interaction.



Agent assist

Empowers agents with real-time prompts and an intelligent document chatbot.



Agent routing

Automatically matches customers to the most suitable agent for their needs.



Channel optimisation

Delivers communications through the most effective channel — SMS, email, app, or call.



Post-call summary

Instantly generates structured notes from call transcripts, saving agents time.



QA automation

Automatically scores every call to ensure consistent quality and compliance.

You must deliver



A working model or agent



Feature Store for ML



An explanation of each decision



Where a human reviews or overrides

EXAMPLE · [Just for reference]

Inside the feature store

features.collections.bank / catalog / hardship_signal

Feature Store

Collections · 142 features · 11 built by LLM

FEATURE CATALOG

Feature	Entity	Type	Source	Refresh	Built by
dpd_max_current	customer	int	gold.c360	daily	SQL
ptp_broken_count_90d	customer	int	promises_to_pay	daily	SQL
payroll_delay_days	customer	int	transactions	daily	SQL
card_utilisation	account	float	card_accounts	daily	SQL
sms_response_rate_30d	customer	float	contact_history	daily	SQL
bureau_score_delta_90d	customer	int	external	monthly	SQL
hardship_signal	customer	enum	notes, calls	per contact	LLM
stated_delay_reason	customer	category	calls	per call	LLM
ptp_intent_strength	promise	0-1	calls	per call	LLM
dispute_flag	account	bool	notes, calls	per contact	LLM
sentiment_last_call	customer	enum	voice → text	per call	LLM

Online vector · G-004817 · 28 Sep 06:00 ET

```
{dpd_max_current: 38, ptp_broken_count_90d: 1,
payroll_delay_days: 29, card_utilisation: 0.94,
sms_response_rate_30d: 0.0, hardship_signal: "clear",
stated_delay_reason: "reduced_income", ptp_intent_strength: 0.55}
```

- 1

SQL + LLM features

Numeric features come from SQL; text and voice features are extracted by an LLM.
- 2

LLM features are tested

Each ships with its prompt, version and accuracy against human labels.
- 3

Same values everywhere

One definition for training (offline) and live scoring (online).

THE DATA YOU'LL GET

Fully synthetic, realistically messy[Few ref datasets]

Dataset	Contents	Format
customers	Profile, segment, consent flags	CSV
card_accounts	Limits, balances, statements, DPD	CSV
loan_accounts	Repayment schedules, instalments, DPD	CSV
deposit_accounts	Balances, salary credits	CSV
collections_cases	Buckets, treatments, promises to pay, outcomes	CSV
contact_history	Channel, time, outcome of each contact	CSV
agent_notes	Free-text notes per contact	Text
call_transcripts	[] synthetic call transcripts	JSON
voice_samples	[] synthetic recordings (optional)	WAV
external	Synthetic bureau-style scores	CSV

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GROUND RULES



Publish a data contract

Data product declares its owner, schema, quality rules and allowed uses.



Explain every decision

A collections agent must understand why your system recommends an action.



Keep a human in the loop

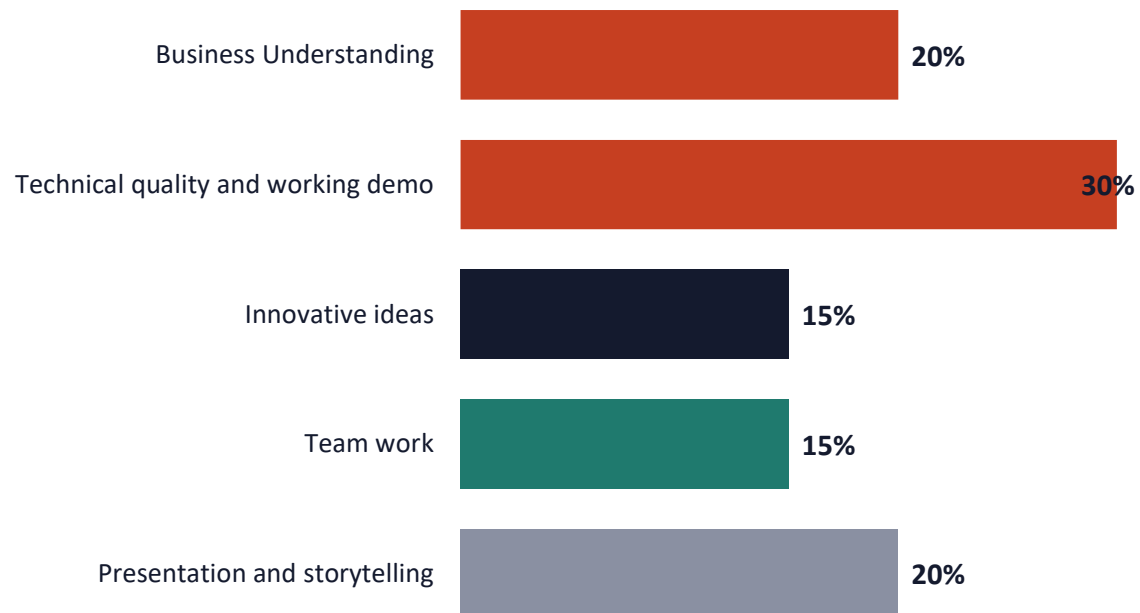
Nothing that affects a customer happens without a person able to review it.



Treat customers fairly

No protected attributes in decisions; flag signs of financial hardship or vulnerability.

How your work will be scored



Submit

-  Code repository
-  Architecture diagram
-  Data contract file
-  5-minute demo video
-  10-slide pitch deck